HTML Quick-Styler– AI Website Generator

Team Members: **N. Nandhini**

V. Guberalakshmi

B.S. Raja Rajeswari

R. Jebaseeli gnanathangam

Abstract

This project presents an AI-assisted web page generator that automatically creates responsive HTML websites based on user input. The system allows users to enter a page title, description of required sections, and select a color style. Using Python and Gradio, the system dynamically generates HTML code and displays a live preview of the website. The goal of the project is to simplify the web development process by enabling users to generate website layouts without writing manual code. The generated pages include common sections such as hero banners, features, pricing, contact forms, and footers with customizable color palettes.

Introduction

Web development requires knowledge of HTML, CSS, and design principles. Many beginners find it difficult to create visually appealing websites from scratch. The purpose of this project is to provide a tool that automatically generates structured website layouts based on simple text input.

The system uses a predefined template engine and color palette generator to build a complete HTML page dynamically. The user interface is created using the Gradio library, which provides an interactive environment for generating and previewing web pages instantly.

This project demonstrates the use of Python programming, UI frameworks, and dynamic web page generation.

Objectives

The main objectives of this project are:

- To develop an automated website generator.
- To allow users to generate HTML pages using simple descriptions.
- To provide multiple color themes for website styling.
- To display the generated webpage through a live preview.
- To simplify web design for beginners.

Scope of the Project

The scope of the project includes generating simple website templates with commonly used sections. Users can customize their websites by selecting different styles and describing the desired layout.

The project is intended for educational purposes and demonstrates how automated web page generation can be implemented using Python.

Future improvements may include:

- AI-based design generation
- Additional website templates
- Database integration
- Website deployment features

System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 MB free space

Software Requirements

- Python 3.x
- Google Colab / Jupyter Notebook
- Gradio library
- Web Browser (Chrome / Edge)

System Architecture

The system works in the following steps:

1. User enters website title, description, and style.
2. The Color Palette Generator selects appropriate colors.
3. The HTML Builder creates the webpage structure.
4. The system generates the full HTML code.
5. Gradio displays a live preview of the website.

Architecture Flow

User Input → Color Palette Generator → HTML Builder → HTML Code Output → Live Preview

Methodology

The project follows these stages:

1. Input Collection

The user provides the title, website description, and preferred color theme.

2. Template Processing

The system analyzes the description and determines which sections to include in the website.

3. HTML Generation

Python functions generate the HTML and CSS structure dynamically.

4. Preview Generation

The generated website is displayed using a Gradio interface.

Implementation

The project is implemented using Python and Gradio.

Key Components

ColorPaletteGenerator Class

- Stores predefined color palettes.
- Returns colors based on selected style.

build_html Function

- Generates the complete HTML structure.

generate_site Function

- Combines user input and templates.
- Produces preview and HTML code.

Gradio Interface

- Provides an interactive UI with tabs:
  - Generate Page
  - Live Preview
  - Color Palettes
  - Templates
  - Guide

Features of the System

- Automatic website generation
- Live preview of generated webpage
- Multiple color themes
- Responsive layout
- Simple user interface
- HTML code output for further customization

Advantages

- Reduces time required to create web layouts
- Easy to use for beginners
- Interactive preview system
- Customizable website sections
- Lightweight and simple implementation

Limitations

- Limited number of website templates
- No database integration
- Requires internet connection for Google Colab
- Advanced animations are not included

Future Enhancements

Future improvements for the system include:

- AI-based layout generation
- More design templates
- Drag-and-drop interface
- Export as downloadable website files
- Integration with hosting platforms

Conclusion

The HTML Quick-Styler Advanced system demonstrates how web pages can be generated dynamically using Python and simple user inputs. The project simplifies website creation by allowing users to describe their required sections instead of writing HTML code manually. The implementation using Gradio provides an interactive interface with live preview capabilities. This project successfully showcases automated web page generation and can be further expanded with advanced features.

References

1. Python Official Documentation
2. Gradio Documentation
3. W3Schools – HTML & CSS Tutorials
4. MDN Web Docs